Customer Lifetime Value (CLV) Analysis Using Probabilistic Modeling Retail Transaction Dataset

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INTRODUCTION AND METHODOLOGY

Purpose of Analysis

The purpose of this analysis is to estimate the long-term economic value of customers based on their historical transaction behavior. By understanding how frequently customers purchase, how recently they last transacted, and how much they typically spend, this analysis aims to support data-driven decisions related to customer segmentation, retention strategy, and resource allocation.

This analysis applies a probabilistic Customer Lifetime Value (CLV) framework using two main components.

First, the BG/NBD (Beta-Geometric / Negative Binomial Distribution) model is used to estimate the probability that a customer is still active and to predict the expected number of future transactions based on frequency, recency, and customer tenure. **Second**, the Gamma-Gamma model is applied to estimate the expected monetary value per transaction for each customer, assuming that purchase frequency and monetary value are independent. The final CLV is calculated by combining the expected future transaction frequency and expected monetary value, adjusted using a discount rate to reflect the time value of money.

Key Conceptual Understanding

Customer Lifetime Value in this analysis represents an **expected value**, not a guaranteed revenue.It reflects both customer behavior uncertainty and the timing of future cash flows, making the result more suitable for realistic business decision-making rather than optimistic revenue forecasting.

BG/NBD → *Will the customer buy again, and how often?*

Gamma-Gamma → *If they buy, how much will they spend?*

Discounting → *Future money is adjusted to today's perspective*

CLV → *Expected future customer value, not a promise*

Modelling Insight

```
Beta_geofitter = BetaGeoFitter(penalizer_coef=0.001)
Beta_geofitter.fit(frequency=df_customer['Frequency'], recency=df_customer['Recency'], T= df_customer['Tenure'])
```

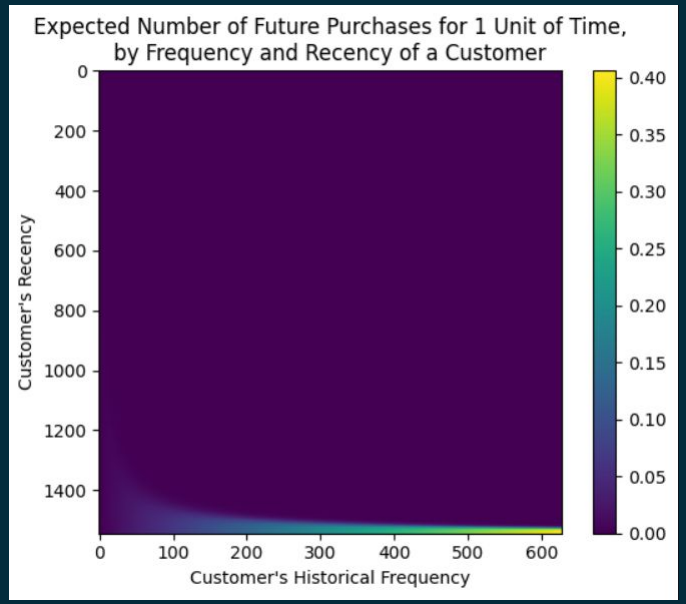
```
<lifetimes.BetaGeoFitter: fitted with 490 subjects, a: 0.45, alpha: 0.02, b: 6.65, r: 0.15>
```

The BG/NBD model was fitted to capture customer purchasing behavior based on frequency, recency, and tenure. The low value of **r (0.15)** indicates a high level of heterogeneity in transaction frequency across customers, meaning that a small group of customers contributes disproportionately to repeat purchases while the majority transact infrequently.

The alpha (0.02) parameter suggests that transactions are generally spaced over longer periods of time, indicating that customer purchases do not occur frequently within short time intervals.

Parameters a (0.45) and b (6.65) describe customer dropout behavior. Since b is significantly larger than a, the model suggests that, on average, customers have a relatively lower probability of churn after each transaction, although individual churn risk still varies across customers. Overall, these parameters support the use of a probabilistic approach to estimate future transaction behavior rather than relying on simple historical averages.

Modelling Insight

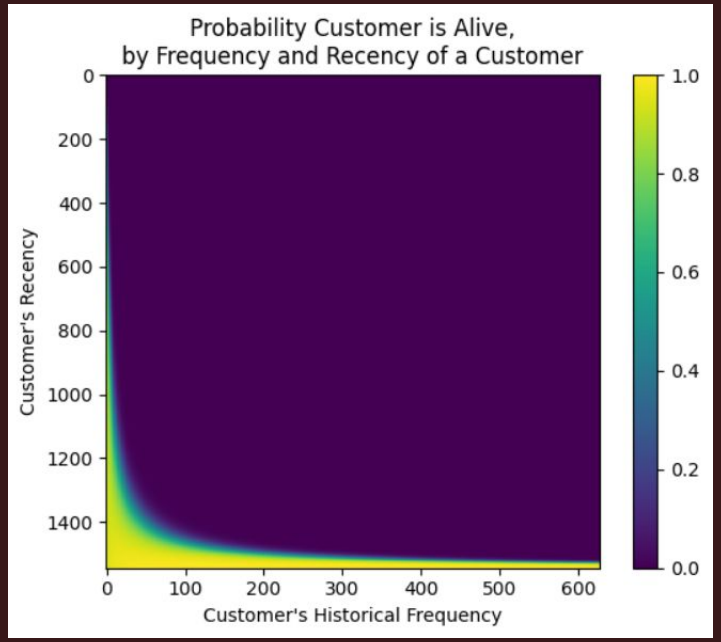


This visualization shows the expected number of future purchases as a function of customer historical frequency and recency. Customers with **high purchase frequency and low recency** (recent transactions) exhibit the highest expected future purchases, indicating a strong likelihood of continued engagement.

Customers with **high frequency but high recency** represent previously active customers who have not purchased recently. Although their historical behavior suggests value, their expected future purchases decline significantly, indicating elevated churn risk.

Customers with **low frequency and high recency** show near-zero expected future purchases and are considered inactive or churned. This analysis suggests that retention strategies should prioritize customers who are both historically active and recently engaged, rather than focusing solely on historical purchase volume.

Modelling Insight



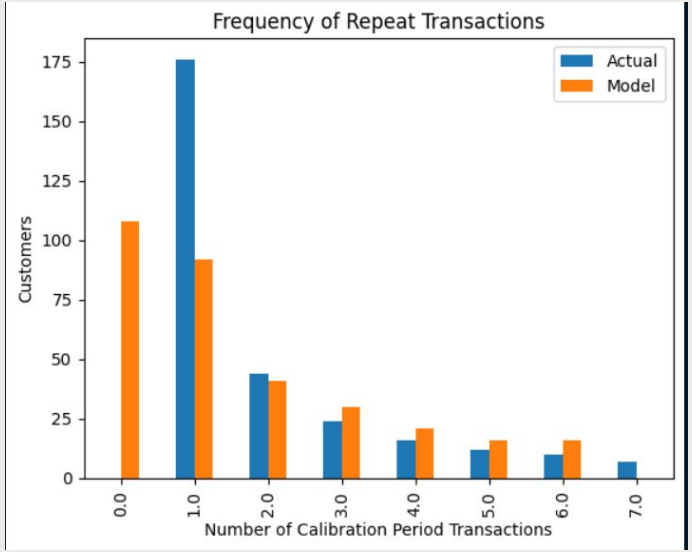
This heatmap illustrates the probability that a customer is still “alive” based on historical purchase frequency and recency.

Customers with **low recency (recent purchases)** consistently exhibit a higher probability of being alive, regardless of their historical frequency.

As recency increases—indicating a longer time since the last purchase—the probability of a customer being alive rapidly declines, even for customers with historically high purchase frequency.

This highlights that **recent engagement is a stronger signal of customer survival than historical activity alone**, and that customers who have not purchased for a long period are highly likely to have churned.

Modelling Insight

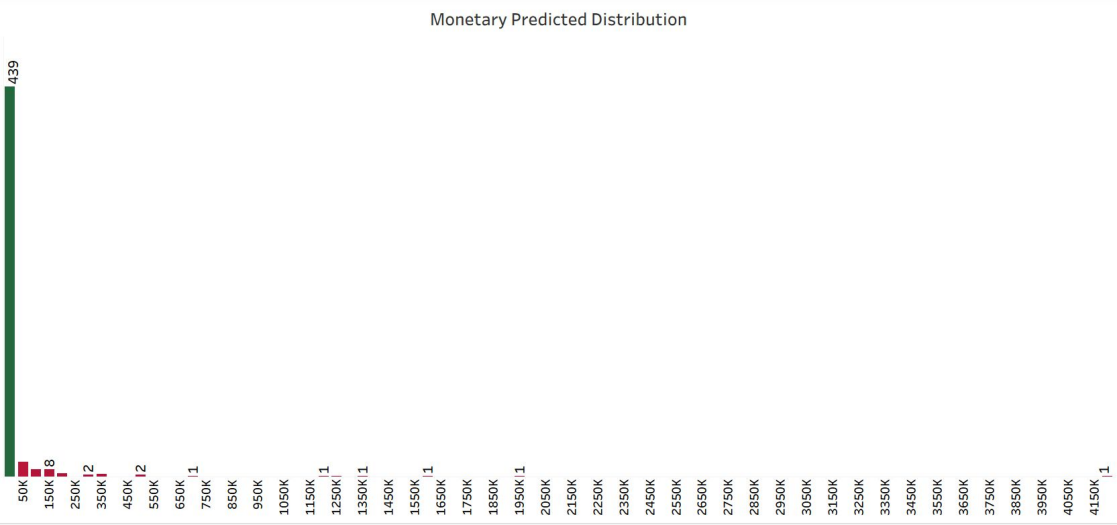


The plot compares the **actual distribution of repeat transactions** during the calibration period with the **distribution predicted by the BG/NBD model**. Overall, the model is able to **capture the general purchasing behavior pattern** of customers. Both the actual and predicted distributions show that most customers make **only one or a small number of repeat transactions**, while the number of customers decreases as transaction frequency increases. This indicates a typical retail pattern where **highly loyal customers are relatively rare**.

However, some differences can be observed. The model slightly **overestimates customers with zero repeat transactions** and **underestimates customers with exactly one repeat transaction**. This suggests that the model assumes a higher probability of early customer inactivity compared to what actually happened in the data. At higher transaction frequencies (three or more purchases), the predicted values follow the actual trend reasonably well, showing that the model performs better in capturing **repeat purchasing behavior among more active customers**.

In summary, this visualization indicates that the BG/NBD model provides a **reasonable fit** to the data and is suitable for estimating future transaction behavior, although minor discrepancies at low transaction frequencies should be considered when interpreting CLV results.

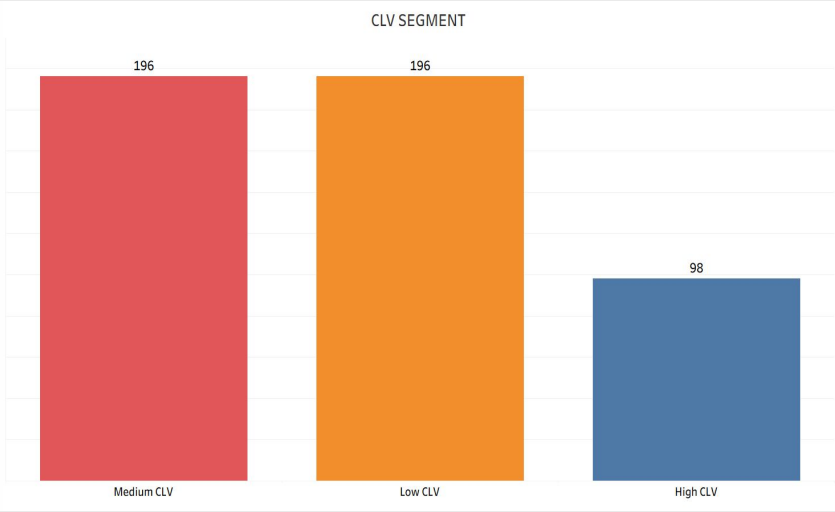
AFTER MODELLING, INSIGHT AND RECOMMENDATION



“The distribution of predicted Customer Lifetime Value shows a highly right-skewed pattern, where the majority of customers contribute relatively low long-term value, while a small group of customers generates disproportionately high value. This indicates that customer value is not evenly distributed, and business performance is strongly influenced by a limited number of high-value customers.”

“From a business perspective, this pattern highlights the importance of prioritizing customer retention over mass acquisition. High-CLV customers should be targeted with personalized retention strategies, loyalty programs, and premium services to protect long-term revenue. Meanwhile, low-CLV customers can be managed using cost-efficient marketing strategies or automated engagement, ensuring resources are allocated efficiently without overspending on low-return segments.”

AFTER MODELLING, INSIGHT AND RECOMMENDATION

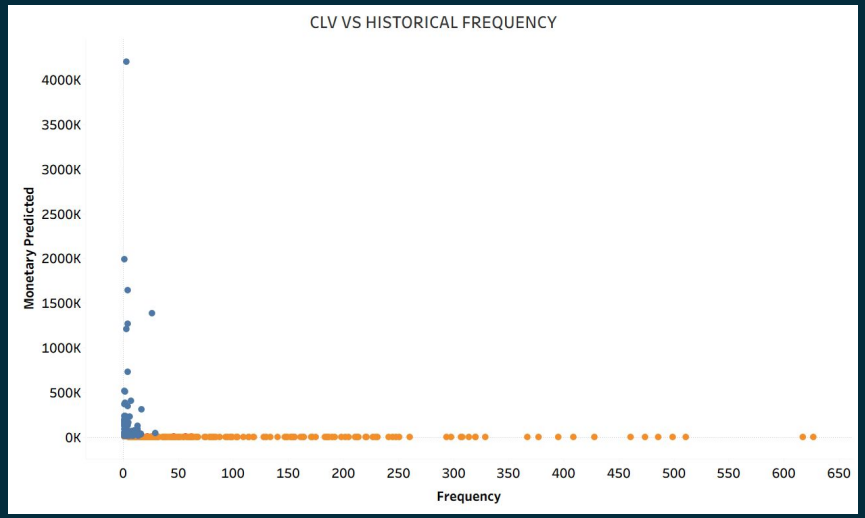


Segmentation is made with percentiles (20 High CLV–40 Medium CLV–40 Low CLV). The CLV segmentation shows that most customers fall into the Medium and Low CLV segments, while High CLV customers represent a relatively smaller portion of the customer base.

This indicates that the business relies on a broad base of customers with moderate to low predicted lifetime value, rather than a large concentration of high-value customers. Such a distribution suggests that long-term revenue sustainability may depend not only on retaining High CLV customers, but also on improving the value contribution of Medium CLV customers over time.

The business should implement a balanced customer strategy by maintaining retention efforts for High CLV customers while placing stronger emphasis on developing Medium CLV customers. Since Medium CLV customers form a large portion of the customer base, targeted initiatives such as personalized promotions and purchase frequency stimulation can gradually increase their lifetime value. This approach reduces dependency on a small High CLV segment and supports more stable long-term revenue growth.

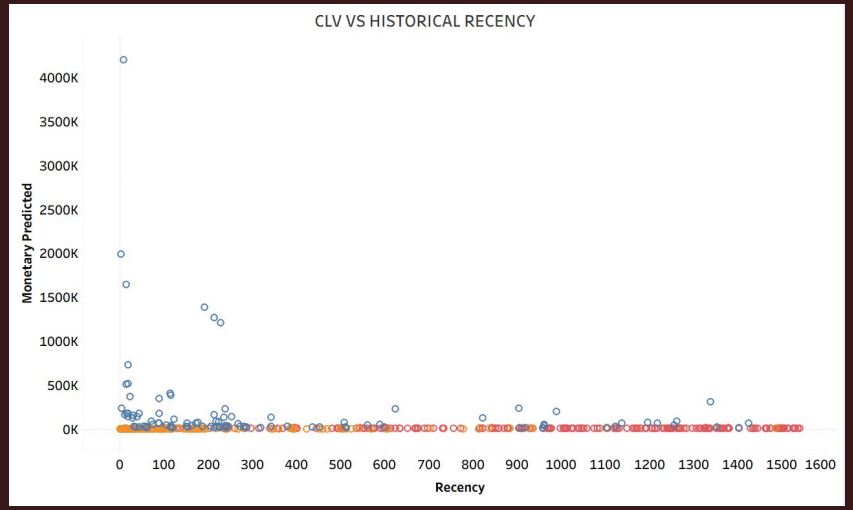
AFTER MODELLING, INSIGHT AND RECOMMENDATION



The visualization shows that higher purchase frequency does not necessarily translate into higher customer lifetime value. High CLV customers tend to have relatively low purchase frequency but generate significantly higher predicted monetary value, indicating that transaction value plays a more important role than transaction count. Conversely, many low CLV customers exhibit high purchase frequency but contribute limited monetary value over time. This pattern suggests that customer value is driven more by quality of transactions rather than quantity.

The business should avoid using purchase frequency as the primary indicator of customer value. Instead, retention and engagement strategies should prioritize customers with high monetary contribution, even if their transaction frequency is low. For high-frequency but low-value customers, strategies should focus on increasing average transaction value through targeted upselling, bundling, or pricing optimization rather than encouraging more frequent purchases.

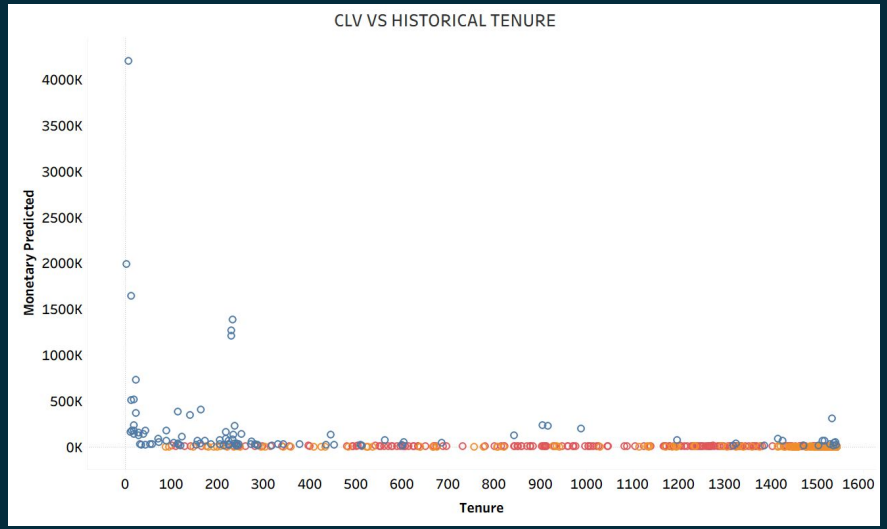
AFTER MODELLING, INSIGHT AND RECOMMENDATION



The visualization indicates that recent purchasing behavior is associated with higher predicted customer lifetime value, but recency alone does not fully explain customer value. While many High CLV customers tend to have low recency, there are also customers with recent transactions who still generate low predicted monetary value. This suggests that customer lifetime value is influenced by a combination of behavioral factors, and recency must be interpreted together with other dimensions such as transaction value and purchasing patterns.

The business should avoid relying solely on recency-based targeting when designing retention strategies. Instead, recency should be combined with monetary contribution to identify truly valuable customers. Recent customers with low predicted value should be approached with strategies focused on increasing transaction value, while High CLV customers with increasing recency should be prioritized for timely re-engagement to prevent potential value loss.

AFTER MODELLING, INSIGHT AND RECOMMENDATION



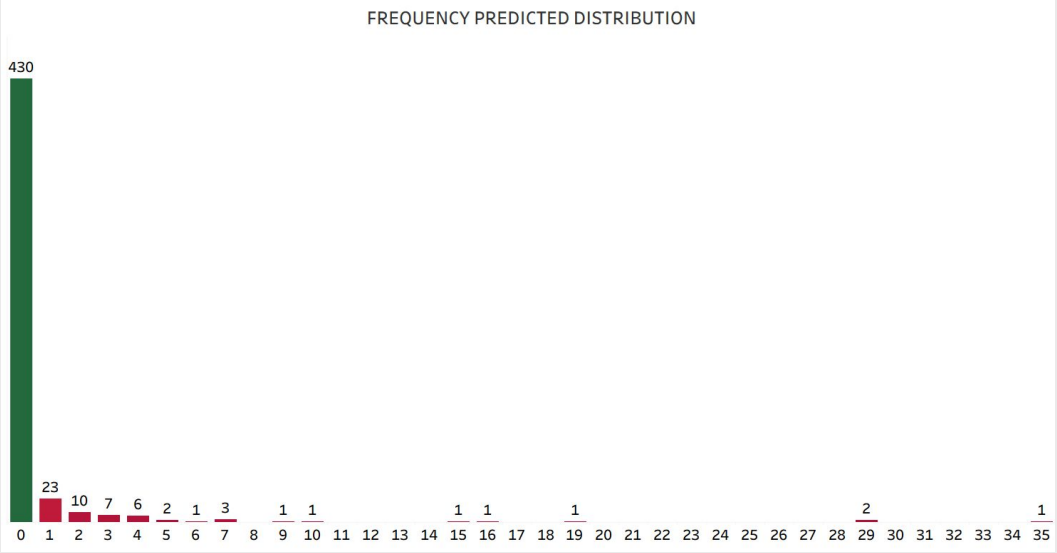
The visualization indicates that longer customer tenure does not necessarily result in higher customer lifetime value. While some High CLV customers appear early in their customer lifecycle, many long-tenured customers remain in the Low and Medium CLV segments with limited predicted monetary contribution. This suggests that customer value is driven more by purchasing behavior and transaction value rather than the length of the customer relationship alone.

Recommendation:

The business should avoid using tenure as the primary indicator of customer value when designing retention strategies. Instead of rewarding customers solely based on how long they have been with the company, retention efforts should focus on customers who demonstrate high monetary contribution. For long-tenured but low-value customers, strategies should prioritize increasing transaction value through targeted offers rather than relying on loyalty duration alone.

AFTER MODELLING, INSIGHT AND RECOMMENDATION

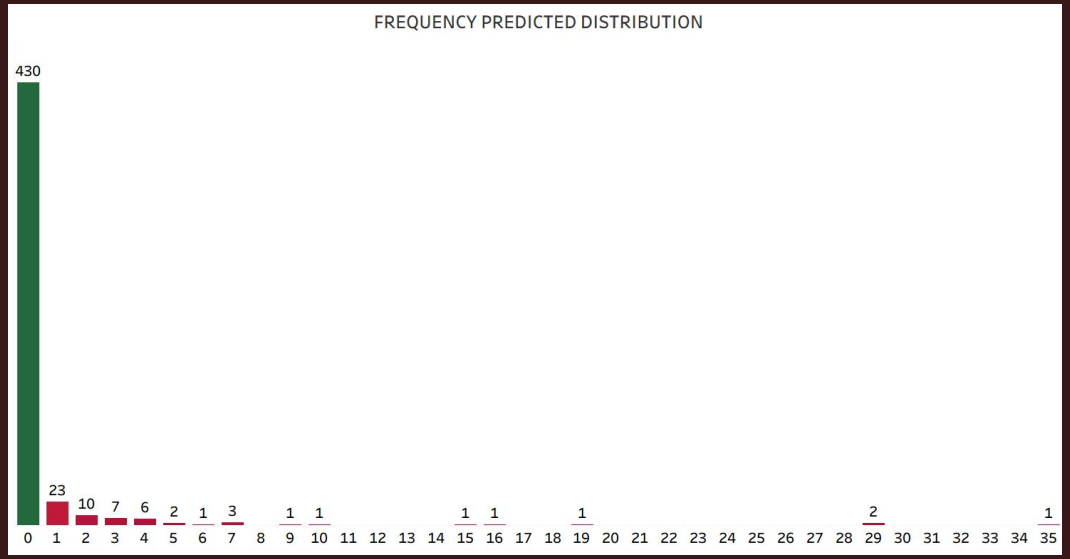
```
df_customer = df_customer[df_customer['Frequency'] > 0]
df_customer = df_customer[df_customer['Recency'] > 0]
```



- Customers with 0 repeat transactions are DISCARDED
- Customers with only one purchase are DISCARDED
- The current dataset contains ONLY repeat customers
- This is a mathematical requirement of BG/NBD, not a business decision.

The model predicts that the customer will NOT make a repeat transaction within the PREDICTION TIME HORIZON.

AFTER MODELLING, INSIGHT AND RECOMMENDATION



Although the dataset consists only of historical repeat customers, the predicted frequency distribution indicates that a large portion of these customers are unlikely to make additional purchases in the future prediction window. This suggests declining engagement or churn risk among historically active customers, with only a small subset expected to continue purchasing.

The business should prioritize retention and personalized engagement strategies for customers with non-zero predicted purchase frequency, as they represent the most reliable source of future revenue. Customers with zero predicted frequency, despite having repeat purchase history, should be considered at risk of churn and targeted with selective reactivation or win-back campaigns, while carefully controlling retention costs.

REFLECTIVE QUESTION

1. Why is data understanding and preprocessing so important before building customer churn and CLV models? Explain how data quality can impact the accuracy of churn predictions and long-term customer value estimates.

Before building a Customer Churn or Customer Lifetime Value (CLV) model, the data understanding and preprocessing stage is critical because these models rely entirely on historical customer behavior patterns. If the input data does not accurately represent real customer activity, the model will learn incorrect signals and produce misleading predictions.

In churn modeling, poor data quality—such as incorrect purchase dates, missing transactions, or inconsistent customer identifiers—can distort key variables like recency, frequency, and tenure. As a result, customers may be incorrectly classified as active or churned, leading to inaccurate churn probabilities. This directly affects business decisions, such as targeting the wrong customers for retention campaigns.

Similarly, in CLV estimation, data issues such as outliers, negative monetary values, or incomplete purchase histories can significantly bias revenue predictions. Since CLV models estimate future value based on past spending behavior, any noise or errors in the monetary data will be amplified over time, causing overestimation or underestimation of customer value.

REFLECTIVE QUESTION

2. What are the business benefits of combining Customer Churn Analysis and Customer Lifetime Value (CLV) Prediction over using either alone? Explain how these two analyses help companies develop more effective customer retention strategies.

Combining Customer Churn Analysis with Customer Lifetime Value (CLV) prediction provides significantly greater business value than using either approach in isolation. While churn analysis identifies *which customers are likely to leave*, CLV estimation explains *how valuable each customer is to the business over time*. When used together, these two analyses enable more precise and economically rational decision-making.

If a company relies only on churn analysis, it may focus retention efforts on customers who are likely to leave but contribute relatively low value. Conversely, if the company uses only CLV prediction, it may identify high-value customers without understanding their risk of churn, leading to missed opportunities for timely intervention.

By combining both analyses, the business can prioritize customers who are **both high-value and high-risk**, ensuring that retention resources such as promotions, loyalty programs, or personalized offers are allocated where they generate the highest return on investment. This approach shifts retention strategy from a volume-based mindset to a value-based one.

In practice, churn analysis answers the question *"Who is likely to leave?"*, while CLV answers *"Who is worth saving?"*. Together, they allow companies to design more effective retention strategies, reduce unnecessary marketing costs, and maximize long-term profitability by focusing on the right customers at the right time.

Thank you

LINK FOR TABLEAU :

https://public.tableau.com/app/profile/magasa.faiz/viz/CLVPREDICTED/FREQUENCYPREDICTEDDISTRIBUTION?publish=yes

LINK FOR JUPYTER NOTEBOOK :

https://drive.google.com/file/d/1g5xrhUPiiD4Ll7bIwjnIsugZK4Nq-pfX/view?usp=sharing

LINK FOR CLV DATA :

https://drive.google.com/file/d/1ZzJmwRDSzJtNE-GfSAe3Q_OwiIZ3FXKK/view?usp=sharing

LINK FOR DATA SCOURCES :

https://www.kaggle.com/datasets/shedai/retail-data-set/data